Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law

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Baseline edition, 12 November 2026

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Acknowledgments and contributions

Kirtus Justus directed the research questions, the selection of mathematical models and the publication standards, is the sole named author, and is responsible for the claims presented here. ChatGPT (OpenAI) assisted with derivations, critical checks and manuscript preparation. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI) and DeepSeek (High-Flyer) contributed to the wider research process; attribution of particular arguments is limited to traceable records. Editorial preparation of this edition: Claude (Anthropic). Assistance does not imply provider endorsement, external peer review or institutional endorsement.

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October 2026

Abstract

The previously derived modular period coefficients can be connected to the finite concentration spectrum with a uniform remainder on every fixed interior interval of the scaled Jacobi spectrum. An exact positive realization of the omitted differential spectrum becomes, after a second change of variable, a smooth whole-line equation with potential bounded away from zero. Its recessive logarithmic derivative gives an exact integral for the concentration logit. An integrable Riccati residual then proves an asymptotic expansion to every fixed finite order. Deformation of the coefficient contours identifies its terms with the original even WKB periods. In particular the expansion through N^{-5} has a uniform $O(N^{-7})$ remainder for $|\lambda/N| \le \tau < 1/2$. A conservative explicit constant is supplied for $\tau = 1/4$. A separate global linear envelope then bounds the positive-majorant-weighted modular discrepancy uniformly. Combined with the established Wilson covariance and boundary-entropy estimates, this gives the fixed-scaled-mass relative entropy $\mu^2 \log L/6 + O_\mu(1)$. Neither a limiting finite part nor a uniform spectral edge expansion is obtained.

1 Statement and conventions

Let $N \ge 2$ be even. For $0 \le j, k < N$, define

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \ne k, \end{cases} \qquad J_{j-1,j} = J_{j,j-1} = \frac{j(N-j)}{N}, \quad J_{jj} = 0. \tag{1}$$

These matrices commute, J has simple spectrum, and $0 < B < I$. For their common modes write

$$Jv = \lambda v, \quad Bv = \zeta v, \qquad t = \lambda/N, \quad \varepsilon = \log\frac{1-\zeta}{\zeta}, \quad e = \varepsilon + N\eta(t), \tag{2}$$

where

$$\eta(t) = \pi t\, {}_3F_2\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 4t^2\right), \qquad |t| < 1/2. \tag{3}$$

The alternating-sign involution sends $(t, \zeta, \varepsilon, e)$ to $(-t, 1-\zeta, -\varepsilon, -e)$. All logarithms are natural.

Theorem 1 (Uniform interior expansion)**.** *For every $0 < \tau < 1/2$ and integer $K \ge 1$, there are finite constants $C_{\tau,K}$ and $N_{\tau,K}$ such that every common mode of (1) with $|t| \le \tau$ and even $N \ge N_{\tau,K}$ satisfies*

$$\left| e - \sum_{j=1}^{K} \frac{\varphi_{2j-1}(t)}{N^{2j-1}} \right| \le C_{\tau,K} N^{-2K-1}. \tag{4}$$

Here the φ_{2j-1} are the even WKB period coefficients defined in Section 5. They are real analytic and odd on $(-1/2, 1/2)$. For $K = 3$ they are precisely the previously reduced functions

$$\varphi_1 = \tfrac{t}{2}\eta' + (t^2 - \tfrac{1}{6})\eta'', \qquad \varphi_3 = A_3\eta' + B_3\eta'', \qquad \varphi_5 = A_5\eta' + B_5\eta'', \tag{5}$$

with the rational coefficients in (39)–(41). *For* $|t| \le 1/4$ *one admissible, deliberately conservative bound is*

$$\left| e - \frac{\varphi_1(t)}{N} - \frac{\varphi_3(t)}{N^3} - \frac{\varphi_5(t)}{N^5} \right| \le \frac{2273823860\,\pi}{N^7}, \qquad N \ge 16. \tag{6}$$

This proves the fixed-interior spectral statement left open in [1]. The bounds apply also across $t = 0$; the apparent singularities there in (5) are removable. The constants depend on the distance from $|t| = 1/2$. No convergence of the full inverse-power series is asserted. The exact identities used from [2, 3] are stated next, so the dependence on those earlier results is explicit.

2 Exact spectral input and a whole-line resolvent

On $I = (-\pi/2, \pi/2)$ let $p(x) = \cos x$ and let

$$A_N = -(pu')' - \frac{N^2 - 1}{4} pu$$

have its Friedrichs realization. Its simple eigenvalues are $\mu_{N,0} < \mu_{N,1} < \cdots$. The first N eigenvalues are $-\operatorname{spec}(NJ/2)$. The remaining spectrum is positive, and the exact tail formula of [2] is

$$F_N(b) = \sum_{k \ge N} \log \frac{\mu_{N,k} + b}{\mu_{N,k} - b} = 2 \sum_{k \ge N} \operatorname{artanh} \frac{b}{\mu_{N,k}}, \tag{7}$$

$$F_N(N\lambda/2) = \log \frac{\zeta}{1 - \zeta}. \tag{8}$$

The series converges locally uniformly for $|b| < \mu_{N,N}$. In [3] the same tail is realized exactly by

$$T_N = -D_z^2 + \frac{4N^2 - 1}{16 p(z)} + \frac{p(z)}{16}, \qquad \frac{\mathrm{d}z}{\mathrm{d}x} = p^{-1/2}, \tag{9}$$

with its Friedrichs realization on the finite z interval. In particular $\mu_{N,N} > N^2/4$ and T_N^{-1} is trace class.

These are operator and boundary-condition statements, not just formal differential substitutions. Briefly, the finite Fourier subspace exhausts the first N modes by a nodal-count argument. The logarithmic derivative of the endpoint connection coefficient is the resolvent trace of A_N. Removing its finite spectral polynomial and integrating that trace from $-b$ to b gives (7)–(8), without an unspecified entire factor. Successive ground-state Darboux deletions realize the omitted spectrum. The transformed Fourier Wronskian is a nonzero constant times $p^{N^2/4}$, yielding (9). At the initial critical endpoint the Friedrichs condition excludes the logarithmic branch; each deletion raises the selected exponent by one. The final endpoints are limit point. These domain arguments are proved in [2, 3]; the algebraic deletion itself belongs to the classical framework of Crum [4].

Undoing the Liouville transformation in (9) gives the unitarily equivalent operator on $L^2(I, \mathrm{d}x)$

$$\widetilde{T}_N = -(pu')' + \left(\frac{N^2}{4p} + \frac{p}{4} \right) u. \tag{10}$$

Indeed its Liouville correction is $p_{xx}/4 - p_x^2/(16p) = -3p/16 - 1/(16p)$.

Introduce the second coordinate

$$r = \log(\sec x + \tan x), \qquad p = \operatorname{sech} r, \qquad \mathrm{d}x = p\,\mathrm{d}r, \qquad h = N^{-1}. \tag{11}$$

For $b = N^2 t/2$, the quadratic form of $\widetilde{T}_N - b$ becomes

$$\int_{\mathbb{R}} \left\{ |u_r|^2 + \left(\frac{N^2}{4} + \frac{p^2}{4} - bp \right) |u|^2 \right\} \mathrm{d}r. \tag{12}$$

It acts in $L^2(\mathbb{R}, p\,\mathrm{d}r)$ and its form domain is $H^1(\mathbb{R})$ when $|t| \le \tau < 1/2$. Define on ordinary $L^2(\mathbb{R}, \mathrm{d}r)$

$$P_h(t) = -h^2 D_r^2 + Q_t + h^2 v, \qquad Q_t = \frac{1 - 2t\,\operatorname{sech} r}{4}, \qquad v = \frac{\operatorname{sech}^2 r}{4}. \tag{13}$$

Put $s_0 = \sqrt{1-2\tau}/2$. Then $Q_t \ge s_0^2 > 0$. All derivatives of $Q_t - 1/4$ and v decay exponentially, uniformly for $|t| \le \tau$.

Lemma 2 (Weighted resolvent identity)**.** *Let $Uu = \sqrt{p}\,u$ map $L^2(p\,\mathrm{d}r)$ unitarily to $L^2(\mathrm{d}r)$. Then*

$$U(\widetilde{T}_N - N^2 t/2)^{-1} U^{-1} = h^2 p^{1/2} P_h(t)^{-1} p^{1/2}. \tag{14}$$

If $G_h(t; r, s)$ is the Green kernel of $P_h(t)$ and $\mathcal{F}_N(t) = F_N(N^2 t/2)$, then

$$\mathcal{F}_N'(t) = \frac{1}{2} \int_{\mathbb{R}} p(r)[G_h(t; r, r) + G_h(-t; r, r)]\,\mathrm{d}r. \tag{15}$$

Proof. The identity follows first on compactly supported tests from (12): the weak equation is $N^2 P_h(t) u = pf$. Coercivity and density extend it to the closed forms and their bounded inverses. This is a weighted resolvent identity; it does not identify the ordinary whole-line spectrum of $P_h(t)$ with the discrete tail spectrum. Positivity and the diagonal bound proved below give $0 < G_h(t; r, r) \le (2hs_0)^{-1}$. Since $\int_{\mathbb{R}} p\,\mathrm{d}r = \pi$, the weighted diagonal is integrable. On compact intervals the positive continuous kernel has trace equal to its diagonal integral; exhaustion gives the same identity on the line. Equivalently, trace class also follows directly from the left side of (14). Differentiating (7) gives $\mathcal{F}_N' = (N^2/2)\,\mathrm{Tr}[(T_N - b)^{-1} + (T_N + b)^{-1}]$. Substitute (14) and $N^2 h^2 = 1$. □

3 An exact Riccati integral for the logit

Let $u_-(r; t, h)$ be the solution of $P_h(t) u = 0$ normalized by $u_-(r) \sim e^{r/(2h)}$ as $r \to -\infty$. Set

$$w = h\frac{u_-'}{u_-}, \qquad hw' + w^2 = Q_t + h^2 v, \qquad w(-\infty) = 1/2. \tag{16}$$

The potential is strictly positive. For precision, w can be constructed by solving (16) from $r = -R$ with value $1/2$ and letting $R \to \infty$. Lower and upper constant barriers give $s_0 \le w \le \max(1/2, \sqrt{\sup(Q_t + h^2 v)})$. The difference of two such solutions satisfies a first-order equation with damping at least $2s_0/h$, giving convergence on compact sets and uniqueness of the bounded solution with the declared left limit. Comparing with $1/2$ gives

$$h(w - 1/2)' + (w + 1/2)(w - 1/2) = -tp/2 + h^2 v. \tag{17}$$

The right side is integrable. The integrating-factor formula therefore shows $w - 1/2 \in L^1(\mathbb{R})$, uniformly on compact t intervals for fixed h, and $w \to 1/2$ at both ends. Reconstruction by integration fixes the required normalization of u_- and proves that it is positive. Define

$$\mathcal{A}_h(t) = \int_{\mathbb{R}} (w(r; t, h) - 1/2)\,\mathrm{d}r. \tag{18}$$

Proposition 3 (Exact connection formula)**.** *For $|t| < 1/2$,*

$$\boxed{\mathcal{F}_N(t) = \frac{\mathcal{A}_h(-t) - \mathcal{A}_h(t)}{h}.} \tag{19}$$

Proof. Difference quotients in t satisfy a damped first-order equation with source $-p/2$. Its integrating-factor formula is dominated in L^1 by an exponential convolution of p. Taking the limit justifies differentiation and gives

$$\partial_t w(r) = -\frac{1}{2h} \int_{-\infty}^{r} p(s) \exp\left[-\frac{2}{h} \int_s^r w(a)\,\mathrm{d}a\right] \mathrm{d}s. \tag{20}$$

Reduction of order constructs the right recessive solution as $u_-(r)\int_r^\infty u_-(a)^{-2}\mathrm{d}a$. Its Wronskian with u_- is -1, so

$$G_h(t;s,s)=\frac{1}{h^2}\int_s^\infty \exp\left[-\frac{2}{h}\int_s^r w(a)\,\mathrm{d}a\right]\mathrm{d}r\le\frac{1}{2hs_0}. \tag{21}$$

Absolute integrability, or Tonelli applied to the nonnegative integrands, permits integration of (20) in r:

$$\mathcal{A}_h'(t)=-\frac{h}{2}\int_{\mathbb{R}} p(s)G_h(t;s,s)\,\mathrm{d}s.$$

The derivative of the right side of (19) is therefore (15). Both sides vanish at $t=0$. This determines the integration constant exactly. □

4 A uniform residual estimate

Smooth positive-potential Liouville–Green expansions and their error bounds are classical [5, 6]. Here an elementary L^1 estimate suffices and keeps the constants tied directly to the exact logit.

For fixed K, recursively choose $a_j(r,t)$ so that

$$S_K=\sqrt{Q_t}\left(1+\sum_{j=1}^K h^{2j}a_j\right), \tag{22}$$

$$\mathcal{R}_K=S_K^2-Q_t-h^2v-\frac{h^2}{2}(\log S_K)''+\frac{h^2}{4}[(\log S_K)']^2 \tag{23}$$

has zero coefficients through h^{2K}. Primes in this section mean r derivatives. At order h^{2j} the new unknown occurs only as $2Q_ta_j$, so the recursion is unique. Every a_j is rational in $p=\operatorname{sech} r$ and t, with possible denominators only powers of $1-2tp$. Moreover $a_j(0,t)=0$ when regarded as a function of p. These facts follow inductively using

$$D_r^2 f=p^2(1-p^2)f_{pp}+p(1-2p^2)f_p,\qquad (D_rf)(D_rg)=p^2(1-p^2)f_pg_p. \tag{24}$$

Thus a_j and every fixed r derivative are $O(p)$ uniformly for $|t|\le\tau$. In particular all integrals below converge.

Choose $h_0>0$ so small that, for $0<h\le h_0$,

$$\sum_{j=1}^K h^{2j}\|a_j\|_\infty\le\tfrac12,\qquad h\left(\tfrac12\|Q_t'/Q_t\|_\infty+2\sum_{j=1}^K h^{2j}\|a_j'\|_\infty\right)\le s_0/2. \tag{25}$$

Here the norms include the supremum over $|t|\le\tau$. They are finite, so such an h_0 exists. Then $S_K\ge s_0/2$ and

$$w_K=S_K-\frac{h}{2}(\log S_K)'\ge s_0/4,\qquad hw_K'+w_K^2-(Q_t+h^2v)=\mathcal{R}_K. \tag{26}$$

Lemma 4 (Integrated stability bound)**.** *Under* (25)*,*

$$\|w-w_K\|_1\le s_0^{-1}\|\mathcal{R}_K\|_1. \tag{27}$$

Consequently

$$\left|\mathcal{F}_N(t)-\frac1h\int_{\mathbb{R}}[S_K(r;-t,h)-S_K(r;t,h)]\,\mathrm{d}r\right|\le\frac{\|\mathcal{R}_K(\cdot,t,h)\|_1+\|\mathcal{R}_K(\cdot,-t,h)\|_1}{hs_0}. \tag{28}$$

Proof. For $d=w-w_K$ the exact equation is

$$hd'+(w+w_K)d=-\mathcal{R}_K,\qquad d(-\infty)=0.$$

The integrating factor and $w + w_K \geq s_0$ yield

$$|d(r)| \leq \frac{1}{h} \int_{-\infty}^{r} |\mathcal{R}_K(s)| e^{-s_0(r-s)/h} \, \mathrm{d}s.$$

Integrating gives (27). Also $S_K \to 1/2$ at both ends, so $\int_{\mathbb{R}} (\log S_K)' \mathrm{d}r = 0$. Substitute (26) into (18) and use (19). □

The cancellation defining a_j is sufficient for an actual uniform integrable residual, not merely a pointwise formal one. Indeed put $z = h^2$ and $A = 1 + \sum_{j=1}^{K} z^j a_j$. If

$$R_0 = -\tfrac{1}{4}(\log Q_t)'' + \tfrac{1}{16}[(\log Q_t)']^2,$$

then direct calculation gives the finite polynomial identity

$$A^2 \mathcal{R}_K = Q_t(A^4 - A^2) + z(R_0 - v)A^2 - \frac{z}{2} AA'' + \frac{3z}{4}(A')^2 + \frac{z}{4}(\log Q_t)' AA'. \tag{29}$$

Its coefficients through z^K vanish. Each remaining coefficient is $O(p)$, uniformly on $0 \leq p \leq 1$, $|t| \leq \tau$; this follows from (24) and $a_j(0,t) = 0$. The denominators are bounded away from zero. Since $A \geq 1/2$, there is a finite $M_{\tau,K}$, given by these finitely many rational functions, for which

$$|\mathcal{R}_K(r,t,h)| \leq M_{\tau,K} p(r) h^{2K+2}, \quad \|\mathcal{R}_K\|_1 \leq \pi M_{\tau,K} h^{2K+2}. \tag{30}$$

Equation (28) now gives a uniform $O(h^{2K+1})$ logit error. The residual formula also provides a direct finite-h bound whenever its norms and the positivity conditions are bounded.

Define

$$\Phi_{2j-1}(t) = \int_{\mathbb{R}} [\sqrt{Q_t}\, a_j(r,t) - \sqrt{Q_{-t}}\, a_j(r,-t)] \, \mathrm{d}r. \tag{31}$$

The uniformly convergent binomial expansion and $\int_{\mathbb{R}} \operatorname{sech}^m r \, \mathrm{d}r = \sqrt{\pi}\, \Gamma(m/2)/\Gamma((m+1)/2)$ give

$$\int_{\mathbb{R}} [\sqrt{Q_{-t}} - \sqrt{Q_t}] \, \mathrm{d}r = \eta(t). \tag{32}$$

It follows that the exact spectral interpolation $\mathfrak{e}_N(t) = N\eta(t) - \mathcal{F}_N(t)$ obeys

$$\mathfrak{e}_N(t) = \sum_{j=1}^{K} \Phi_{2j-1}(t) N^{1-2j} + O_{\tau,K}(N^{-2K-1}) \tag{33}$$

on the whole interval $[-\tau, \tau]$. The integrals in (31) are odd and real analytic there. At a common finite mode, (8) identifies $\mathfrak{e}_N(t)$ with e.

5 Identification with the original periods

For completeness, the finite Fourier equation for an eigenmode is

$$(\cos\xi\, U')' + \left[\frac{N^2 - 1}{4} \cos\xi - \frac{N\lambda}{2}\right] U = 0.$$

Writing $u = \cos\xi$ and $V = \sqrt{u}\, U$ gives

$$V'' + [N^2 q(u,t) + r_0(u)] V = 0, \qquad q = \frac{u - 2t}{4u}, \qquad r_0 = \frac{1}{4u^2}. \tag{34}$$

The formal even momentum is

$$W = N\sqrt{q} \left(1 + \sum_{j \geq 1} N^{-2j} f_j(u,t)\right), \qquad W^2 + \tfrac{1}{2}(\log W)_{\xi\xi} - \tfrac{1}{4}[(\log W)_\xi]^2 = N^2 q + r_0. \tag{35}$$

For $0 < |t| < 1/2$, let Γ be counterclockwise around the cut joining 0 and $2t$, excluding ± 1, and put

$$\varphi_{2j-1} = i \oint_\Gamma \sqrt{q}\, f_j \, \mathrm{d}\xi, \qquad \mathrm{d}\xi = -\frac{\mathrm{d}u}{\sqrt{1-u^2}}. \tag{36}$$

The branch of $\sqrt{1-u^2}$ is positive on $(-1,1)$ with cuts on the outer real rays; $\sqrt{q} \sim 1/2$ at infinity and has its cut between 0 and $2t$. These conventions agree with [1].

Proposition 5 (Exact equality of coefficient integrals)**.** *For every $j \geq 1$ and $|t| < 1/2$, $\Phi_{2j-1}(t) = \varphi_{2j-1}(t)$, with the latter extended at zero.*

Proof. In (35) set $\xi = ir$, so that $u = \cosh r = 1/p$. Second derivatives and products of first derivatives change sign, while $q = Q_t$ and $r_0 = v$. The resulting even-momentum equation is exactly (23) with its residual set formally to zero. Uniqueness of the coefficient recursion gives

$$a_j(p,t) = f_j(1/p, t). \tag{37}$$

In particular $f_j(u,t) = O(1/u)$ at infinity. It is rational, with possible poles only at 0 and $2t$, and it satisfies $f_j(-u,t) = f_j(u,-t)$.

For $j \geq 1$ the differential in (36) is $O(u^{-2})\mathrm{d}u$ at infinity. Thus Γ can be deformed to the two outer branch cuts: the circle at infinity contributes zero, and the small arcs about ± 1 vanish because their singularities have order $1/2$. On the upper bank of the positive outer cut, $\sqrt{1-u^2} = -i\sqrt{u^2-1}$; on the upper bank of the negative cut it is $+i\sqrt{u^2-1}$. The lower values have opposite signs. With the stated counterclockwise orientation this yields

$$\varphi_{2j-1} = 2\int_1^\infty \frac{\sqrt{q(u,t)}f_j(u,t) - \sqrt{q(u,-t)}f_j(u,-t)}{\sqrt{u^2-1}}\, \mathrm{d}u. \tag{38}$$

Substitute $u = \cosh r$ and use evenness in r to obtain (31). The analytic continuation through $t = 0$ follows from that uniformly convergent real integral. The leading coefficient is handled separately by (32); the unsubtracted leading differential does not have the decay needed for this contour deformation. □

Together, (33), Proposition 5 and (8) prove (4). For the first three functions the exact rational reductions of [1] give (5), with

$$A_3 = \frac{108t^4 - 83t^2 + 7}{120t(4t^2-1)^2}, \qquad B_3 = \frac{708t^6 - 789t^4 + 153t^2 - 14}{360t^2(4t^2-1)^2}, \tag{39}$$

$$A_5 = -\frac{\begin{array}{c}6384t^{10} - 51320t^8 + 26567t^6 \\ -13594t^4 + 2019t^2 - 124\end{array}}{1680t^3(4t^2-1)^4}, \tag{40}$$

$$B_5 = -\frac{\begin{array}{c}38304t^{12} - 374688t^{10} + 267522t^8 \\ -170573t^6 + 39472t^4 - 4937t^2 + 248\end{array}}{5040t^4(4t^2-1)^4}. \tag{41}$$

These formulae were proved by exact derivatives on the period curve; the present argument supplies their spectral connection and remainder. Their small-t behavior includes $\varphi_1(t) = \pi t/4 + O(t^3)$, consistent with the separately proved plunge-law sign $e \simeq -\varepsilon/(4N^2)$ in this convention. The absolute uniform bound above alone is not a relative error bound for arbitrarily small nonzero t.

Corollary 6 (Interior spectral projection)**.** *Let $\Pi_{N,\tau} = \mathbf{1}_{[-\tau,\tau]}(J/N)$ and $E_N = \log(I-B) - \log B + N\eta(J/N)$. Under the conditions of Theorem 1,*

$$\left\| \Pi_{N,\tau} \left[E_N - \sum_{j=1}^{K} N^{1-2j} \varphi_{2j-1}(J/N) \right] \Pi_{N,\tau} \right\| \leq C_{\tau,K} N^{-2K-1}.$$

Proof. Apply (4) in the common orthonormal eigenbasis. □

6 Global envelope and weighted consequences

The fixed-interior expansion has a useful global complement. Write $K_N = \log(I-B) - \log B$, and retain $E_N = K_N + N\eta(J/N)$.

Theorem 7 (Global linear bounds)**.** *For every even $N \geq 2$, in the common reference functional calculus,*

$$\frac{2\pi}{3}|J| \leq |K_N| \leq 2\pi|J|, \qquad |E_N| \leq 2\pi|J| \leq \pi N I. \tag{42}$$

Proof. Take $0 \leq t < 1/2$ and $h \leq 1/2$ in the exact Riccati construction. Write $w_+ = w(r;t,h)$, $w_- = w(r;-t,h)$, and $d = w_- - w_+$. The potentials lie between zero and $9/16$, so constant barriers give $0 < w_\pm \leq 3/4$. Moreover $Q_{-t} + h^2 v \geq 1/4$ gives $w_- \geq 1/2$. Subtraction yields

$$hd' + (w_- + w_+)d = tp, \qquad d(-\infty) = 0.$$

Thus $d \geq 0$, $d \in L^1$, and d tends to zero at both ends. Integrating, and using $1/2 \leq w_- + w_+ \leq 3/2$, gives

$$\frac{2\pi t}{3} \leq \int_{\mathbb{R}} d\,\mathrm{d}r \leq 2\pi t.$$

Proposition 3 converts this to $2\pi N t/3 \leq \mathcal{F}_N(t) \leq 2\pi N t$. Oddness handles negative t and (8) gives the first two bounds in (42). Also

$$0 \leq \sqrt{Q_{-t}} - \sqrt{Q_t} = \frac{tp}{\sqrt{Q_{-t}} + \sqrt{Q_t}} \leq \sqrt{2}\,tp,$$

since $Q_{-t} + Q_t = 1/2$. Therefore $0 \leq \eta(t) \leq \sqrt{2}\pi t$. The two nonnegative quantities $N\eta(t)$ and $\mathcal{F}_N(t)$ are each at most $2\pi N t$, so their difference has absolute value at most $2\pi N t$. Finally $\|J\| \leq N/2$. □

Theorem 8 (Positive weighted modular control)**.** *Suppose $G_N \geq 0$ and constants $M, a, b \geq 0$, independent of N, satisfy*

$$\operatorname{Tr} G_N \leq M, \qquad \nu_N((x,\infty)) := \operatorname{Tr}[G_N \mathbf{1}_{\{|J|>x\}}] \leq a/x + b/x^2 \quad (x > 0). \tag{43}$$

Then

$$\operatorname{Tr}(G_N|E_N|) = O_{M,a,b}(1). \tag{44}$$

More precisely, there are universal C, N_0 such that

$$\operatorname{Tr}(G_N|E_N|) \leq CM/N + 4\pi a + 16\pi b/N \quad (N \geq N_0). \tag{45}$$

For the reference occupation weight $W_N = B(I-B)$, independently,

$$\operatorname{Tr}(W_N E_N^2) = O\left(\frac{\log N}{N^2}\right). \tag{46}$$

Proof. Theorem 1 with $K = 1$ and $\tau = 1/4$ gives $|e| \leq C/N$ on $|\lambda| \leq N/4$. The complementary part is bounded by $\pi N \nu_N((N/4,\infty))$ using (42). This proves (45). Only nonnegative diagonal expectations of G_N are used; G_N need not commute with the reference matrices.

For (46), the exact kernel calculation gives

$$\operatorname{Tr} W_N = \frac{2N}{\pi^2} \sum_{\substack{d \geq N \\ d \text{ odd}}} \frac{1}{d^2} + \frac{2}{\pi^2} \sum_{\substack{1 \leq d < N \\ d \text{ odd}}} \frac{1}{d} = O(\log N).$$

On the interior, $e^2 \leq C^2/N^2$. On the complement, (42) gives

$$\zeta(1-\zeta) = \frac{1}{4\cosh^2(|\varepsilon|/2)} \leq e^{-|\varepsilon|} \leq e^{-\pi N/6}, \qquad e^2 \leq \pi^2 N^2.$$

Summing at most N modes bounds that contribution by $\pi^2 N^3 e^{-\pi N/6}$, proving (46). □

Application to the specified Wilson interval

The application is to the occupied-band projector of the infinite chain

$$h_m(k) = \sin k\,\sigma_x + (1 + m - \cos k)\sigma_z, \qquad P_m(k) = \tfrac{1}{2}[I - h_m(k)/|h_m(k)|],$$

compressed to L consecutive two-orbital cells, with $N = 2L$, $m = \mu/L$, and fixed $\mu > 0$. Let A_N, B_N denote the massive and massless compressed covariances after the interval-preserving scalarization. The previously proved decomposition $A_N - B_N = S_N + D_N^{\mathrm{tr}}$ has [7]

$$-G_N \le S_N \le G_N, \qquad M = \mu/\pi, \quad a = \sqrt{2/15}\,\mu^2/\pi, \quad b = \mu/(3\pi), \tag{47}$$

$$|\langle v_j, D_N^{\mathrm{tr}} v_j\rangle| \le 2\sqrt{\langle v_j, G_N v_j\rangle\,\zeta_j(1-\zeta_j)}. \tag{48}$$

The constants in (47) are those of (43). They follow from the positive Fourier representation of G_N and exact Jacobi bond sums, not a spectral approximation. In particular Theorem 8 implies

$$|\operatorname{Tr}(S_N E_N)| = O_\mu(1), \qquad |\operatorname{Tr}(D_N^{\mathrm{tr}} E_N)| \le 2\sqrt{M \operatorname{Tr}(W_N E_N^2)} = O_\mu\left(\frac{\sqrt{\log N}}{N}\right). \tag{49}$$

Corollary 9 (Leading Wilson relative entropy)**.** *Use the Wilson covariance definitions above and the separately established bounded nonlinear and boundary-entropy estimates of [8]. For the relative entropy with the massive covariance in the first argument,*

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + O_\mu(1) \quad (L \to \infty,\ \mu > 0 \textit{ fixed}). \tag{50}$$

Proof. The exact finite identity is

$$\mathcal{D}_L = \mathcal{P}_L + \mathcal{Q}_N + \operatorname{Tr}[(S_N + D_N^{\mathrm{tr}})E_N] - \Delta S_L,$$

where $\mathcal{P}_L = \operatorname{Tr}[(A_N - B_N)(-\pi J)]$, $\mathcal{Q}_N = \operatorname{Tr}[(A_N - B_N)(-N\eta(J/N) + \pi J)]$, and $\Delta S_L = S(A_N) - S(B_N)$. The imported estimates used here are precisely

$$\mathcal{P}_L = \frac{\mu^2}{6}\log L + O_\mu(1), \qquad \mathcal{Q}_N = O_\mu(1), \qquad \Delta S_L = O_\mu(1).$$

The first is the exact two-bond elliptic asymptotic; the second uses the positive-majorant fourth moment and the reference occupation-Jacobi moment; the third is the cross-boundary trace-class entropy estimate. These bounded statements are proved separately in [8], independently of its stronger conditional finite-part hypotheses. Substitution of (49) proves (50). □

This corollary imports those specified Wilson theorems; the present paper supplies the previously missing modular estimate. It does not assume the conditional all-degree convergence, Coulomb comparison, or entropy finite-part convergence statements in [8]. Boundedness of the remainder in (50) does not imply that it has a limit.

The positive-trace formulation is consistent as well. Put $Q_N = \operatorname{Tr}(G_N|J|)$ and $P_N = \operatorname{Tr}(G_N|K_N|)$. The positive-coefficient series for η gives $0 \le |\eta(t)| - \pi|t| \le C_\eta|t|^3$ on $|t| \le 1/2$, where $C_\eta = 16\,\mathrm{arsinh}(1) - 4\pi$. Layer-cake integration of (43) gives

$$0 \le \operatorname{Tr}[G_N N|\eta(J/N)|] - \pi Q_N \le C_\eta(3a/8 + 3b/(2N)).$$

Since the scalar difference between $|K_N|$ and $N|\eta(J/N)|$ has absolute value $|E_N|$, Theorem 8 proves

$$P_N = \pi Q_N + O_\mu(1) = \frac{\mu^2}{6}\log N + O_\mu(\sqrt{\log N}). \tag{51}$$

The second equality follows without a separate sharp Jacobi-moment input. The independent positive-trace reduction in [9] is $\mathcal{D}_L = P_N + O_\mu(\sqrt{\log N})$. Corollary 9, proved directly from the four branches without P_N or Q_N, therefore yields the displayed asymptotic for P_N. The already proved $P_N = \pi Q_N + O_\mu(1)$ then gives the corresponding coefficient of Q_N. There is no use of either positive-trace asymptotic in the proof of Corollary 9.

7 Numerical checks and remaining scope

The proof does not use fitted constants or numerical mode values. As a separate check of its exact normalization, set $z = e^r$ and $u_- = z^{N/2} f(z)$. Equation (13) becomes

$$z(1+z^2)^2 f'' + (N+1)(1+z^2)^2 f' + [N^2 t(1+z^2) - z] f = 0, \qquad f(0) = 1. \tag{52}$$

Frobenius expansion followed by overlapping Taylor expansions computes $f(1), f'(1)$. Evenness of the potential gives the growing connection coefficient

$$\mathfrak{a}_h(t) = \frac{2f(1)[Nf(1)/2 + f'(1)]}{N}, \qquad \mathcal{F}_N(t) = \log \mathfrak{a}_h(-t) - \log \mathfrak{a}_h(t).$$

Here $\log \mathfrak{a}_h = \mathcal{A}_h/h$, as follows by integrating $w - 1/2$ over the line. This computation uses the whole-line equation, whereas the retained reference values were computed from finite Jacobi eigenvectors and concentration quadratic forms. For four $N = 160$ modes, at t approximately $-0.04909, -0.12511, -0.25111, -0.40225$, the absolute differences in the logits are respectively 3.90×10^{-89}, 7.61×10^{-90}, 2.27×10^{-88} and 4.78×10^{-88}. This is consistent with the retained input precision. Separately, direct 110-digit finite-matrix calculations at every mode for $N = 2, 4, 8, 16, 32, 64$ satisfy both global logit bounds, the discrepancy envelope, and the particle–hole sign convention.

Exact symbolic substitution verifies (37) for $j = 1, 2, 3$ and the four vanishing coefficients of (29) below h^8. Real quadrature of (31) at six points between -0.3 and 0.45 agrees with (5) to absolute errors below 6×10^{-66}. A further check evaluates the exact interpolation at fixed t; these points need not be finite Jacobi eigenvalues. Table 1 displays the scaled remainder after the first three terms.

N	$t = 0.125$	$t = 0.3$	$t = 0.45$
16	-2.756381	-286.9294	-2.020035×10^6
32	-2.854713	-327.4185	-4.221313×10^6
64	-2.881067	-340.4583	-6.240675×10^6
128	-2.887778	-343.9819	-7.261610×10^6

Table 1: $N^7[\mathfrak{e}_N(t) - \varphi_1(t)/N - \varphi_3(t)/N^3 - \varphi_5(t)/N^5]$ from the exact whole-line interpolation. The much larger values near $t = 1/2$ illustrate the importance of fixing the interior interval.

All decimal checks use multiprecision arithmetic with increased precision and truncation for replication; they are not outward-rounded interval enclosures. The constant in (6), by contrast, comes from exact rational inequalities in the appendix. It is much larger than the observed errors and is not presented as a sharp bound.

The result resolves the fixed-interior period connection and remainder problem for the half-filled sine kernel. It leaves open uniform control in a moving edge window $1/2 - |t| \to 0$. Consequently the earlier coefficient-level Stirling limits do not yet become a uniform spectral edge theorem. The bound on the interior projection does not by itself control $\|E_N\|$ on the full spectrum. The separate global envelope and Wilson spectral tail suffice for the leading entropy law, but do not establish a limiting finite part or the proposed inverse-logarithmic rate. General filling and the continuum prolate problem require separate exact operators and domain analysis; they are not covered by Theorem 1. Within *From Prephysical Hypotheses to Physical Models*, these are dimensionless reference-model results. They neither select a physical clock nor establish a universal physical model or a quantum-field-theoretic mass gap.

A A finite rational certificate for the explicit constant

Here $K=3$, $|t|\le 1/4$, $p\in[0,1]$, $z=h^2$ and $A=1+za+z^2b+z^3c$. The first three coefficients can be generated by

$$\begin{aligned}
&R_0=-\tfrac14(\log Q)''+\tfrac{1}{16}[(\log Q)']^2, && a=(v-R_0)/(2Q),\\
&R_1=-\tfrac12 a''+\tfrac14(\log Q)'a', && b=-(Qa^2+R_1)/(2Q),\\
&L_2=b-a^2/2, \qquad R_2=-\tfrac12 L_2''+\tfrac14(\log Q)'L_2'+\tfrac14(a')^2, && c=-(2Qab+R_2)/(2Q).
\end{aligned}$$

Each $a_j=n_j(p,t)/(1-2pt)^{3j}$ for a polynomial n_j divisible by p. For example

$$a=\frac{p(p^3t^2-pt^2+p-2t)}{2(1-2pt)^3}.$$

For a polynomial $n=\sum n_{ab}p^at^b$, the rational number $\sum|n_{ab}|4^{-b}$ bounds $|n|$ on this whole rectangle. Applying this to n_j/p, n_j, and its first two p derivatives, and using $1-2pt\ge 1/2$, gives the following uniform majorants:

j	$\lvert a_j\rvert/p\le M_j$	$\lvert D_r a_j\rvert/p\le D_j$	$\lvert D_r^2 a_j\rvert/p\le E_j$
1	13/2	31	379/2
2	7353/8	17595/2	386379/4
3	6923583/16	49698847/8	1566168253/16

For example, with $m=3j$, $\delta=1/2$ and the indicated polynomial norms,

$$\begin{aligned}
\|\partial_p a_j\|_\infty &\le \|n_j'\|\delta^{-m}+2m\tau\|n_j\|\delta^{-m-1},\\
\|\partial_p^2 a_j\|_\infty &\le \|n_j''\|\delta^{-m}+4m\tau\|n_j'\|\delta^{-m-1}+4m(m+1)\tau^2\|n_j\|\delta^{-m-2}.
\end{aligned}$$

Equation (24) then bounds D_j by the first expression and E_j by the sum of these two expressions.

Take $h_0=1/16$ and replace s_0 conservatively by $s_*=1/3$. The two left sides of (25) are bounded, respectively, by

$$\frac{17504063}{268435456}<\frac12, \qquad \frac{117523487}{1073741824}<\frac16=\frac{s_*}{2}.$$

Thus $A\ge 1/2$ and the stability argument applies for every $h\le 1/16$.

To specify the residual bound without a long expanded polynomial, let $M(z)=1+\sum_{j=1}^3 M_jz^j$, $D(z)=\sum_{j=1}^3 D_jz^j$, and $E(z)=\sum_{j=1}^3 E_jz^j$. Write $[z^k]$ for coefficient extraction. Since $Q\le 3/8$, $|R_0-v|/p\le 13/16$, and $|(\log Q)'|/p\le 1$, the coefficient of z^k in the right side of (29), for $4\le k\le 12$, has absolute value at most pC_k, where

$$C_k=[z^k]\left\{\tfrac38(M^4+M^2)+z\left[\tfrac{13}{16}M^2+\tfrac12 ME+\tfrac34 D^2+\tfrac14 MD\right]\right\}. \tag{53}$$

Every product in these positive-power coefficients contains a factor vanishing at $p=0$, so the single factor p is retained. The coefficients for $k<4$ vanish exactly by the recurrence. Since $A^{-2}\le 4$,

$$\|\mathcal{R}_3\|_1\le 4\pi h^8\sum_{k=4}^{12}C_kh_0^{2k-8}.$$

The two signs in (28) and $s_*=1/3$ give a multiplier $24\pi\sum_{k=4}^{12}C_k16^{-2k+8}$. Exact rational evaluation of this finite expression yields

$$24\sum_{k=4}^{12}C_k16^{-2k+8}<2273823860.$$

This proves (6) using finite rational inequalities.

Acknowledgments and computational assistance. ChatGPT (OpenAI) assisted with the present analysis, symbolic identities, numerical implementations and proof scrutiny. The earlier endpoint connection proposal was supplied through work with Claude (Anthropic); the precise identities and assumptions used here are recorded in [2, 3]. The broader program has also used Gemini (Google), Grok (SpaceXAI) and DeepSeek (High-Flyer). This does not attribute the present checks to those systems or imply endorsement. The mathematical argument and reproduction materials remain open to independent review; AI-assisted checking is not independent human peer review.

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